$$\mathcal{N}_S : Stm \to Stm$$

$$\mathcal{N}_E : Exp \to \overline{Stmt}, Exp$$

- $\mathcal{N}_E(x) = x$

- $\mathcal{N}_E(v) = v$

- $$\frac{\mathcal{N}_E(e) = stmts, e' \qquad x \text{ fresh}}{\mathcal{N}_E(\oplus e) = (stmts; x = e'), x}$$

$$\frac{N_E(e_1) = stmts_1,\ e_1' \qquad N_E(e_2) = stmts_2,\ e_2' \qquad x \text{ fresh}}{N_E(e_1 \oplus e_2) = (stmts_1;\ stmts_2;\ x = e_1' \oplus e_2'),\ x}$$

$$\frac{N_E(e) = stmts,\ e' \qquad N_E(e_i) = stmts_i,\ e_i' \Big|_{i=1,\dots,n} \qquad x \text{ fresh}}{N_E(e(e_1,\dots,e_n)) = stmts;\ stmts_1;\ \dots\ stmts_n;\ x = e'(e_1',\dots,e_n'),\ x}$$

$$\frac{N_E(e) = stmts_e,\ e' \qquad N_S(s_i) = s_i' \Big|_{i=1,2}}{N_{stmt}(\text{if}(e)\ \{s_1\}\ \text{else}\ \{s_2\}) = stmts_e;\ \text{if}(e')\ \{s_1'\}\ \text{else}\ \{s_2'\}}$$